Aetna Live Chat Support System — AWS Serverless Architecture

AWS Well-Architected Framework · 🔐 Security · 🔁 Reliability · ⚡ Performance · ⚙️ Operational Excellence · 💰 Cost Optimization · 🌱 Sustainability · 🏥 HIPAA-Aligned

👤 USER LAYER Web · Mobile

- 🌐 React Chat Widget (Aetna.com)
- 📱 React Native Mobile App (iOS + Android)
- WSS + HTTPS protocols
- Hosted on S3 + CloudFront

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🛡️ EDGE + SECURITY LAYER 🔐 SEC ⚡ PERF

☁️ Amazon CloudFront

TLS 1.3 termination · Custom domain chat.aetna.com
Geo-restrict to US · ACM cert auto-renew
Cache: static 1yr TTL · API 5min TTL

🔥 AWS WAF

OWASP Top 10 · SQL Injection · XSS · LFI/RFI
Rate limit: 100 req/min per IP · Bot Control
AWS Shield Standard (auto) / Advanced (opt)

🔑 Amazon Cognito

Member Pool + Agent Pool (separate)
OIDC → Aetna IdP federation · JWT RS256 · 1h TTL
MFA: TOTP/SMS · RBAC: member/agent/supervisor/admin

🔐 Security Scope

L1 Edge: DDoS + WAF
L2 Identity: OIDC + JWT + MFA
L3 Network: TLS 1.3 + HSTS header enforced

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🔀 API GATEWAY LAYER 🔐 SEC 🔁 REL ⚡ PERF

⚡ WebSocket API

de>$connect → JWT validate + session create
de>sendMessage → route to BOT or AGENT
de>$disconnect → cleanup + flush events
de>typing → typing indicator UX

🌐 HTTP REST API

de>GET /health · de>POST /v1/chat/start
de>GET /v1/chat/history
de>GET /v1/agents/status
de>/v1/admin/* (supervisor panel)

- Cognito JWT Authorizer (all routes)
- Throttle: 10K RPS burst · 5K steady
- Usage plans per consumer
- Access logging → CloudWatch

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⚙️ LAMBDA COMPUTE LAYER ⚡ PERF 🔁 REL ⚙️ OPS

- connect-fn · 256MB · 512ms
- router-fn · 256MB · 1s
- disconnect-fn · 128MB · 256ms
- chatbot-fn · 512MB · 3s
- agent-router-fn · 256MB · 1s
- escalation-fn · 256MB · 1s
- action-server-fn · 512MB · 5s
- analytics-fn · 256MB · 2s
- audit-fn · 128MB · 1s
- Provisioned concurrency (hot paths)
- X-Ray tracing on all functions
- VPC private subnets only

Least-privilege IAM per function | RDS Proxy → Aurora connection pool | SQS DLQ on all async invocations

CHATBOT ENGINE

Amazon Lex V2 50+ intents · NLU · Entity extraction · Slot filling · Confidence scoring

Amazon Kendra KB index · FAQ · Semantic search · 100K documents

Confidence ≥ 0.70 → BOT resolves

0.40–0.70 → BOT responds + offer escalation

< 0.40 or "agent" keyword → escalate

AGENT ROUTING

Amazon SQS FIFO Skill-based queue · Priority tiers · DLQ 14d retain

Scoring: skill_match + (1−load) + resolution_rate + recency

SNS / SES Real-time agent notify · Email · SMS alerts

Agent presence via Redis · Load-balanced assignment

Agent UI: S3+CloudFront SPA + WebSocket API

EVENT BUS

REL | OPS

Amazon EventBridge

Topics: chat.started · chat.message · chat.escalated · chat.resolved

Rules route to: analytics-fn · audit-fn · Kinesis Firehose

DLQ on every rule for zero message loss

Kinesis Firehose

Streams all events → S3 (raw log archive)

Buffer: 128MB or 60s (whichever first)

Delivers Parquet/JSON → Athena queryable

DATA LAYER

SEC | REL

DynamoDB

Sessions + Messages

On-demand capacity

Multi-AZ by default

PITR 35 days

KMS CMK encrypted

TTL: session expiry

Aurora PostgreSQL

Members + Reports

Multi-AZ + replicas

35d automated PITR

TDE + KMS encrypted

RDS Proxy pooling

Aurora Serverless v2

ElastiCache Redis

Session context cache

Member profile cache

Cluster mode, multi-AZ

Sub-5ms latency

TTL: 5min member data

Agent presence store

S3 + Glacier

Audit logs (WORM)

Object Lock enabled

AES-256 + KMS CMK

7-year retention

Lifecycle: hot→IA 30d → Glacier 90d

ANALYTICS

OPS | COST

Amazon Redshift Chat aggregates · Agent performance · Trend analysis

Amazon Athena Ad-hoc SQL on S3 · Log analysis · Cost attribution

Amazon QuickSight Executive dashboards · Resolution rate · Agent KPIs · CSAT

Dialogflow interaction logs → BigQuery equivalent in Redshift

OBSERVABILITY

OPS | REL

CloudWatch Metrics + Alarms p50 <100ms · p95 <500ms · p99 <2s · Error <0.5%

AWS X-Ray Distributed tracing · Service map · Bottleneck detection

CloudWatch Logs Structured JSON · Log Insights · 30d hot · S3 archive

PagerDuty On-call rotation · P1/P2/P3 runbooks · Post-incident reviews

🔒 SECURITY INFRASTRUCTURE — ALL 8 LEVELS 🔐 SEC

Level	Layer	AWS Service	Controls
L1 Edge	Perimeter	CloudFront + Shield + WAF	DDoS mitigation · OWASP rules · Geo-restriction · TLS 1.3
L2 Identity	Auth	Cognito + Aetna IdP	OIDC federation · JWT RS256 · MFA · RBAC roles
L3 Network	VPC	VPC + SGs + NACLs	Private subnets · VPC endpoints · No public DB · Flow logs
L4 Application	Code	Lambda IAM + API GW	Least-privilege per fn · Input validation · No wildcard policies
L5 Transit	In-flight data	ACM + TLS + WSS	TLS 1.3 enforced · HSTS header · cert auto-renew · WSS only
L6 Storage	At-rest data	KMS CMK + AES-256	Separate CMK per service · 90d rotation · Field-level PHI encrypt
L7 Audit	Detection	CloudTrail + GuardDuty + Macie	Immutable logs · ML anomaly detection · PHI/PII auto-discovery
L8 Compliance	Governance	Security Hub + Config + BAAs	HIPAA control mapping · Drift detect · BAAs signed with AWS

🚀 CI/CD PIPELINE ⚙️ OPS 🔁 REL

GitHub → **CodePipeline** → **CodeBuild** (SAST+DAST+unit tests) → **SAM/CDK Deploy** → **Staging** → **Canary 10%→50%→100%** → **Production**

Auto-rollback: error rate >5% or p99 >2s triggers immediate rollback · Blue/green Lambda aliases · AppConfig feature flags

🔁 RESILIENCE / DISASTER RECOVERY 🔁 REL RTO < 5 min RPO < 5 min

Multi-AZ Primary (us-east-1)

Lambda + API GW: multi-AZ auto

Aurora Multi-AZ: failover <30s

DynamoDB: multi-AZ by default

ElastiCache: cluster mode multi-AZ

SQS DLQ: retain 14 days

Multi-Region Standby (us-west-2)

Aurora Global DB replica

DynamoDB Global Tables

S3 Cross-Region Replication

Route53 health check failover

Warm standby pattern · GameDay quarterly

🏥 **HIPAA Alignment:** §164.312(a)(1) Access Control · §164.312(b) Audit Controls · §164.312(c)(1) Integrity · §164.312(d) Authentication · §164.312(e)(1) Transmission Security · §164.312(e)(2)(ii) Encryption at Rest · Breach Notification (60-day) · BAA with AWS signed

User / Edge · Security · API Gateway · Compute / Lambda · Chatbot Engine · Agent Routing · Event Bus · Data Layer · Analytics · Observability · Security Infra · Resilience / DR